

Decant Groceries and Label SOP

One episode decants one bulk pack of dry groceries into one jar. The pack is opened, poured into the jar up to the fill line, and thrown out, then the jar is closed with its lid, labeled on its body, and stood in the crate.

The order never changes: stage, open, pour, discard the pack, close, label, shelve. The pack is not lifted over the jar until its rolled top is unrolled and its pack mouth stands open, the lid does not go on until the pour is finished, the label does not go on until the jar is closed, and the jar does not go in the crate until it is closed and labeled.

The left gripper stages the jar, holds the label sheet flat, and steadies the jar while the right gripper works on it. The right gripper takes the pack, pours, discards the empty pack, closes the lid, applies the label, and shelves the finished jar. Neither gripper crosses to the other side of the table.

The jar is never lifted while it is open and holding product, nothing is turned in the air, no gripper opens anywhere but over the spot the thing in it is going to, and product that leaves the pack goes into the jar and nowhere else.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual
- **parked_side:** none

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the pack spot at the back right, the crate at the back center, the jar spot at the front left, the lid spot right of the working area, the label sheet in front of the working area, and the waste bin at the front right corner.
3. The working area at the center of the table is visible from above, so the pour can be seen going into the jar mouth and any product that lands on the table can be seen.
4. Both arms are at home with grippers open.
5. The table is bare apart from the pack, the jar, the lid, the label sheet, the crate, and the waste bin.
6. The left arm reaches the jar spot, the working area, and the label sheet. The right arm reaches the pack spot, the lid spot, the label sheet, the working area, the crate, and the

waste bin.

Materials checklist

1. One **bulk pack** of dry groceries stands at the back right, closed by a rolled top. It holds one jar fill or less, so the whole pack empties into the jar without reaching the fill line.
2. One empty **jar** stands upright at the front left, mouth open and base flat, carrying a printed **fill line** on its body below the jar mouth.
3. One **lid** for that jar lies on the table right of the working area, interior down against the clean table, so the lid is lifted straight onto the jar mouth and never has to be turned over.
4. One **label sheet** lies flat on the table in front of the working area, written side up, with no corner curled. It carries one written **label** naming the product in the pack.
5. One **crate** with four slots in a row sits at the back center, slots running left to right, each deep enough that a jar stands in it without leaning. At least one slot is open. Slots already holding a closed and labeled jar from a previous episode stay as they are, and the open slots run to the right of the filled ones.
6. One open **waste bin**, empty, at the front right corner, its rim clear of the label sheet.
7. The working area at the center of the table is clear and dry.

Workspace layout

- **Pack spot:** back right, the closed bulk pack standing on the table.
- **Crate:** back center, four slots in a row running left to right, at least one of them open.
- **Jar spot:** front left, the empty jar standing upright.
- **Working area:** the center of the table, where the open, the pour, the close, and the label happen.
- **Lid spot:** right of the working area, the lid lying interior down on the table.
- **Label sheet:** in front of the working area, lying flat on the table, written side up.
- **Waste bin:** front right corner, open and empty.

Everything the right arm works on is staged on the right or in the center. The jar, the only thing the left arm carries, is staged on the left. No carried thing crosses the midline.

Vocabulary

- **Arm assignment:** the left gripper stages the jar, holds the label sheet flat on the table, and steadies the jar while the right gripper works on it. The right gripper takes the pack, pours, discards the empty pack, closes the lid, applies the label, and shelves the jar. Neither gripper takes the other's zones.
- **Bulk pack:** one flexible pack of dry groceries, closed by a rolled top, holding one jar fill or less. Its **pack base** is the closed bottom end, opposite the rolled top.

- **Rolled top:** the top of the bulk pack folded over on itself to hold it closed.
- **Pack mouth:** the opening of the bulk pack once the rolled top is unrolled and its two sides are pulled apart.
- **Product:** the dry groceries inside the bulk pack. Product goes from the pack into the jar and nowhere else.
- **Jar:** the rigid storage jar that receives the product from the pack. Its **jar mouth** is the threaded opening at the top and its **jar body** is the straight side wall between the base and the jar mouth.
- **Fill line:** the printed mark on the jar body below the jar mouth. It is the highest the product may sit.
- **Lid:** the screw lid that closes the jar. It is held only by its **lid rim**, the outer edge. A gripper never closes on the interior of the lid, and the lid is never turned over in the air.
- **Closed jar:** the lid turned onto the jar mouth until it stops turning, sitting level with no gap at the jar mouth.
- **Turn and regrip:** how the lid is closed. Turn the lid clockwise as far as the wrist goes, open the gripper, bring the wrist back, close on the lid rim again, and turn again. Repeat until the lid stops turning. The lid is never closed in one continuous twist.
- **Label sheet:** the backing sheet carrying the written label, lying flat on the table in front of the working area. It is held down by the left gripper for the peel and is never lifted.
- **Label:** the written adhesive label naming the product. Its **label face** is the written side.
- **Labeled jar:** a closed jar carrying the label pressed flat on the jar body, label face out and the writing upright.
- **Crate slot:** one of the four upright positions in the crate that holds one jar.
- **Steady hold:** the left gripper closed on the jar body with the jar upright and its base flat on the table, held for the whole of the pour, the close, or the label press.
- **Lift:** coming straight down onto the grasp point, closing, and raising straight up until there is daylight under the thing, before anything moves sideways.
- **Lower in:** carrying level over the crate slot or the waste bin, lowering straight down until the thing is resting, and releasing once it is resting. Nothing is dropped in from above the rim.
- **Settled:** the thing stays put for 2 seconds after the gripper lifts clear, with nothing rocking, sliding, or tipping.
- **Release point:** over the working area, over the crate slot, or over the waste bin the thing in the gripper is going to. A gripper opens nowhere else.
- **Spill:** any product outside the bulk pack, the jar, or the waste bin.

Steps

Steps 1 to 7 fill the jar from the pack and run once. Step 8 ends the episode once the jar is in the crate.

Step 1: Stage the jar and the pack in the working area

Goal: the empty jar standing upright in the working area with the closed pack lying flat beside it on its right.

- The **left gripper** closes on the **jar body** of the jar at the jar spot, lifts it straight up, carries it upright to the **working area**, lowers it until its base is flat, and releases.
- The **right gripper** closes on the **bulk pack** at the pack spot around its middle, lifts it straight up, carries it level to the working area, lays it flat beside the jar on the jar's right with the rolled top pointing toward the jar, and releases.

Each gripper carries one thing at a time, and neither one drags its thing across the table.

Check: the jar stands upright in the working area with its base flat and its mouth open, and the closed pack lies flat beside it, clear of it, with the rolled top pointing toward the jar.

Step 2: Open the bulk pack

Goal: the rolled top unrolled and the pack mouth standing open, with the pack still flat on the table.

- The **left gripper** closes on the **pack base** and holds the pack flat on the table.
- The **right gripper** closes on the outer edge of the **rolled top** and unrolls it away from the pack base until the top lies flat and the pack mouth is clear.
- If the rolled top does not come free in one unroll, take the outer edge again and unroll a second time.
- The **right gripper** then closes on one side of the **pack mouth** and pulls it away from the other side until the mouth stands open.
- Both grippers release.

The pack stays flat on the table for the whole of this step. Never lift the pack to open it, and never lift the closed pack over the jar.

Check: the rolled top lies flat, the pack mouth stands open, the pack is still flat on the table, and no product has come out of it.

Step 3: Decant the pack into the jar

Goal: all of the product in the jar at or below the fill line and the pack empty.

- The **left gripper** takes a **steady hold** on the jar body.
- The **right gripper** closes on the **pack base**, lifts the pack clear of the table, and brings the pack mouth over the jar mouth, centered on it and just above it.
- Tilt the pack toward the jar mouth in one slow motion and hold the pack mouth centered over the jar mouth while the product runs in.

- If the product stops running while product is still in the pack, bring the pack upright, then tilt again.
- Once no product is left in the pack, bring the pack upright and hold it clear of the jar. The **left gripper** keeps its steady hold until the pack is clear.

The jar stays on the table through the whole pour, so never lift the jar to meet the pack and never pour from a height above the jar mouth. If the product reaches the fill line while product is still in the pack, stop pouring and end the episode as an overfilled pack (see Failures that are not violations).

Check: the pack is empty, the product sits at or below the fill line, and there is no product on the table, in the crate, or anywhere outside the jar.

Step 4: Discard the empty pack

Goal: the empty pack in the waste bin and the working area holding only the open jar.

- The **right gripper**, still holding the empty pack, carries it level to the **waste bin**, lowers it in, and releases.
- The **left gripper** releases the jar once the pack is clear of the working area.

Check: the waste bin holds the empty pack, the working area holds only the open jar standing upright, and no product came out of the pack on the way.

Step 5: Close the jar

Goal: a closed jar, its lid seated level and turned until it stops.

- The **left gripper** takes a **steady hold** on the jar body.
- The **right gripper** closes on the **lid rim** of the lid at the lid spot, lifts it straight up, carries it level over the jar, and lowers it onto the **jar mouth**, seated square and sitting level.
- Close it with **turn and regrip**, repeating until the lid stops turning.
- Release the right gripper, then the left.

Hold the lid by its rim only. The interior of the lid touches the product, so no gripper closes on it and the lid is never turned over in the air. Never lift the jar while it is open.

Check: the lid sits level on the jar mouth with no gap, it does not turn further, and it does not lift off when the jar body is nudged.

Step 6: Label the jar

Goal: the label pressed flat on the jar body, naming the product in the jar.

- The **left gripper** closes on the near corner of the **label sheet** and holds the sheet flat on the table.

- The **right gripper** closes on the corner of the **label** and peels it off the sheet in one motion with the label face out.
- The **left gripper** releases the sheet and takes a **steady hold** on the jar body, jar upright, while the right gripper holds the label clear of the table.
- Bring the label to the side of the **jar body** facing the front edge, label face out and the writing upright.
- Press it onto the jar body centered on the jar body height, then press along it from one edge to the other until it lies flat with no lifted corner.
- Release both grippers.

The sheet stays flat on the table for the whole of the peel. Read the label face before peeling it and confirm it names the product just decanted. If it names something else, or if it tears or folds onto itself, end the episode (see Failures that are not violations).

Check: the jar is closed and carries the label, the label face names the product in the jar, the writing is upright, the label lies flat with no lifted corner, the label face points toward the front edge, and the label sheet still lies flat on the table.

Step 7: Shelf the jar in the crate

Goal: the closed and labeled jar standing upright in the leftmost open crate slot with its label face toward the front of the table.

- The **right gripper** closes on the **jar body** without turning the jar, lifts it straight up, carries it upright to the **crate**, and lowers it into the leftmost open **crate slot** until its base sits flat.
- Release once the jar is resting.

The jar is carried in the orientation it was labeled in, so the label face still points toward the front of the table when it is set down. Nothing is turned in the air. Never press the jar down into a slot and never lower it into a slot that already holds one.

Check: the jar stands upright in the leftmost open crate slot with its base flat, it does not lean, and its label face points toward the front of the table.

Step 8: Confirm the jar and end the episode

- Confirm the jar stands closed and labeled in the crate with its label face toward the front of the table and the jars shelved in earlier episodes untouched, and confirm the pack spot, the jar spot, and the lid spot are empty, the waste bin holds the empty pack, and the working area is bare.
- Return both arms home, clear of the crate and the working area, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Leave the shelved jar in its crate slot. It stays there while the crate fills over the following episodes.
2. If all four slots are now full, lift the four jars out, open each one counterclockwise and set its lid aside, empty the product back into fresh bulk packs, peel the label off each jar body, and set the empty crate back at the back center with all four slots clear.
3. Stand one fresh closed pack at the back right, its rolled top rolled and holding one jar fill or less.
4. Stand one empty jar upright at the front left, mouth open and base flat. Replace a jar whose fill line has worn off or whose thread will not take a lid.
5. Wipe the lid clean and lay it interior down right of the working area.
6. Lay a fresh label sheet in front of the working area, written side up and flat with no corner curled, carrying one label that names the product in the pack.
7. Empty the waste bin and set it back at the front right corner.
8. Wipe any product off the table and confirm the surface is clean and dry. A crumbed table lets the jar slide under a steady hold.
9. Vary which product is staged between episodes rather than running the same one every time. The count stays at one pack, one jar, one lid, and one label.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** the pack is lifted over the jar before its pack mouth is open, the lid goes on before the pour is finished, the label goes on before the jar is closed, the jar goes in the crate before it is labeled, or the empty pack is still in the working area when the lid goes on.
- **SOP rule broken:** Steps 1 to 7, the order never changes.
- **Coaching note:** stage, open, pour, discard, close, label, shelve. Finish the step you are in before you start the next one.

Violation: Arms swapped roles

- **Visible cue:** the left gripper takes the pack, the lid, the label off the sheet, or carries the jar to the crate or the waste bin; or the right gripper takes the jar from the jar spot.
- **SOP rule broken:** Steps 1 to 7, the left gripper stages and steadies the jar and holds the label sheet, and the right gripper does the pack, the lid, the label, and the shelving.
- **Coaching note:** each arm stays on its own side. Nothing crosses the midline.

Violation: Jar and pack handled in one grasp

- **Visible cue:** a gripper closes on the jar and the pack together, or lifts the pack while it is still carrying the jar.
- **SOP rule broken:** Step 1, each gripper carries one thing at a time.
- **Coaching note:** one jar, one pack, one label, one lid, one slot. Set one down before you take the next.

Violation: Pack lifted to open it

- **Visible cue:** the pack leaves the table during the unroll, the rolled top is worked open in the air, or the pack mouth is pulled open while the pack hangs from a gripper.
- **SOP rule broken:** Step 2, the pack stays flat on the table for the whole of the open.
- **Coaching note:** hold the pack base down and unroll against the table. Nothing opens in the air.

Violation: Poured while the pack was still closed

- **Visible cue:** the pack is lifted over the jar with the rolled top still rolled, or with the pack mouth never pulled open.
- **SOP rule broken:** Steps 2 and 3, the pack mouth stands open before the pack goes over the jar.
- **Coaching note:** open it flat on the table, then lift it. Never carry a closed pack over the jar.

Violation: Jar not steadied during a pour, close, or press

- **Visible cue:** the left gripper is off the jar body while the product runs, while the lid is turned, or while the label is pressed; or the jar tips, slides, or lifts off the table during any of them.
- **SOP rule broken:** Steps 3, 5, and 6, the left gripper holds the jar upright with its base flat for the whole of each.
- **Coaching note:** the steady hold goes on before the right gripper starts and comes off after it finishes.

Violation: Poured from too high or off center

- **Visible cue:** the pack mouth is well above the jar mouth or is not over it when the tilt starts, or it drifts off the jar mouth while the product runs.

- **SOP rule broken:** Step 3, bring the pack mouth just above the jar mouth and centered on it, and hold it there for the whole pour.
- **Coaching note:** get low and centered before you tilt, and stay there until it stops running.

Violation: Spill

- **Visible cue:** product lands on the table, in the crate, on the label sheet, or anywhere outside the jar and the waste bin at any point in the episode.
- **SOP rule broken:** Step 3, all of the product goes into the jar.
- **Coaching note:** center the pack mouth and tilt in one slow motion so the product runs rather than drops.

Violation: Kept pouring past the fill line

- **Visible cue:** the product sits above the fill line at the end of the pour, or the tilt continues after the product has reached it.
- **SOP rule broken:** Step 3, the product sits at or below the fill line.
- **Coaching note:** watch the fill line during the pour and bring the pack upright as soon as the product reaches it.

Violation: Product left in the pack

- **Visible cue:** the pack is set down or thrown out with product still in it while the product in the jar is below the fill line.
- **SOP rule broken:** Step 3, the pack is empty at the end of the pour.
- **Coaching note:** bring it upright and tilt again until nothing more runs out.

Violation: Empty pack not discarded

- **Visible cue:** the lid goes on, the label goes on, or the jar is shelved while the empty pack is still in the working area, at the pack spot, or in the crate.
- **SOP rule broken:** Step 4, the empty pack goes in the waste bin before the jar is closed.
- **Coaching note:** the pack goes straight from the pour to the bin. Do not put it down first.

Violation: Lid mishandled

- **Visible cue:** a gripper closes on the interior of the lid instead of the lid rim, or the lid is turned over between the lid spot and the jar mouth.
- **SOP rule broken:** Step 5, the lid is held by its rim only and is never turned over in the air.
- **Coaching note:** the interior touches the product. Take the rim and carry it the way it lies.

Violation: Jar not closed

- **Visible cue:** the lid rests on the jar mouth without being turned, sits at an angle, shows a gap, or lifts off when the jar is moved.

- **SOP rule broken:** Step 5, seat the lid level and turn and regrip until it stops turning.
- **Coaching note:** seat it square first, then turn and regrip. One twist is never enough.

Violation: Lid twisted without regripping

- **Visible cue:** the wrist runs to its limit and stalls while still closed on the lid, or the gripper drags on the lid rim instead of opening and taking a fresh hold.
- **SOP rule broken:** Step 5, open the gripper, bring the wrist back, and close on the lid rim again between turns.
- **Coaching note:** turn, let go, come back, take it again. The wrist does not have the range to close a lid in one go.

Violation: Label sheet not held down

- **Visible cue:** the left gripper is off the label sheet during the peel, or the sheet lifts, slides, or folds while the right gripper pulls the label off it.
- **SOP rule broken:** Step 6, the left gripper holds the sheet flat on the table for the whole of the peel.
- **Coaching note:** pin the sheet first, then peel. The sheet never leaves the table.

Violation: Label crooked, off center, or lifting

- **Visible cue:** the writing on the label face is not upright, the label sits high or low on the jar body rather than centered on its height, or a corner of it stands off the jar.
- **SOP rule broken:** Step 6, the label goes on with the writing upright, centered on the jar body height, and pressed flat with no lifted corner.
- **Coaching note:** line the writing up before it touches, then press from one edge to the other.

Violation: Label applied to an open jar

- **Visible cue:** the label is peeled or pressed while the jar is not yet a closed jar.
- **SOP rule broken:** Steps 5 and 6, the jar is closed before it is labeled.
- **Coaching note:** close it, then label it. Never the other way round.

Violation: Jar shelved in the wrong slot

- **Visible cue:** the jar goes into a crate slot with an open slot to its left, or into a slot that already holds a jar.
- **SOP rule broken:** Step 7, the jar goes into the leftmost open crate slot.
- **Coaching note:** find the leftmost open slot before you lower the jar.

Violation: Jar shelved leaning, pressed down, or off zone

- **Visible cue:** the jar is released outside the crate, left in the working area, set down leaning, or pressed down into a slot after it is already resting.

- **SOP rule broken:** Step 7, lower the jar into the slot until its base sits flat and release once it is resting.
- **Coaching note:** take it all the way down, then let go. Leave it how it lands.

Violation: Label face turned away from the front

- **Visible cue:** the jar is turned in the gripper between the label press and the crate, or the label face on the shelved jar points to the back or the side of the table.
- **SOP rule broken:** Step 7, the jar is carried in the orientation it was labeled in and nothing is turned in the air.
- **Coaching note:** the orientation is set when the label goes on. Carry it straight back and do not twist.

Violation: Jar lifted while open

- **Visible cue:** the jar leaves the table between the pour and the close, or the open jar holding product is carried, tipped, or slid.
- **SOP rule broken:** Steps 3 to 5, the jar is never lifted while it is open and holding product.
- **Coaching note:** the jar stays on the table until its lid is on.

Violation: Dragged instead of lifted

- **Visible cue:** the jar, the pack, or the lid slides along the table still touching it, with no daylight under it in transit.
- **SOP rule broken:** Steps 1 to 7, lift straight up until there is daylight before moving sideways.
- **Coaching note:** the lift is its own motion, not the start of the carry.

Violation: Regripped in the air

- **Visible cue:** the gripper shifts, rolls, or seats its hold again on the jar, the pack, or the label without setting it down first. The turn and regrip cycle in Step 5 is not this.
- **SOP rule broken:** Steps 1 to 7, set it down before taking a new hold.
- **Coaching note:** put it down before you take a new hold.

Violation: Dropped instead of lowered

- **Visible cue:** a gripper opens above the crate or the waste bin and the thing falls the rest of the way in, or the jar is released before its base is resting.
- **SOP rule broken:** Steps 1, 4, and 7, release only once the thing is resting.
- **Coaching note:** take it all the way down before you open the gripper.

Violation: Gripper released over the wrong place

- **Visible cue:** a gripper opens over the table, over the crate when it is carrying the pack, over the waste bin when it is carrying the jar, or over open space.
- **SOP rule broken:** Steps 1 to 7, a gripper opens over the spot the thing is going to and nowhere else.
- **Coaching note:** know where it lands before you open the gripper.

Violation: Placement corrected by shoving

- **Visible cue:** a gripper nudges, pats, or slides the jar, the pack, the lid, or the label into position after releasing it, instead of lifting it clear and placing it again.
- **SOP rule broken:** Steps 1 to 7, correct a bad placement by lifting the thing clear and placing it again.
- **Coaching note:** lift it and lay it again, or leave it alone.

Violation: Repeated fiddling

- **Visible cue:** more than about two small corrections to settle the jar, the pack, the lid, or the label rather than settling it in one or two.
- **SOP rule broken:** Steps 1 to 7, each placement settles in one or two small corrections.
- **Coaching note:** the approach is off, so the first placement lands wide. Tighten the approach rather than correcting after.

Violation: Dropped, collided, or knocked something over

- **Visible cue:** the jar, the pack, the lid, or the label falls to the table or the floor short of its spot, a jar in the crate is knocked over, the label sheet or the waste bin is struck and shifts, or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 7, keep each thing on a clear travel path and release only after a settled placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with the jar out of the crate, unlabeled, or open, the pack still on the table, the lid still at the lid spot, product on the table, an arm away from home, or an arm still over the crate or the working area.
- **SOP rule broken:** Step 8, confirm the jar and the empty zones; return both arms home clear of the scene; then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Overfilled pack:** the product reaches the fill line while product is still in the pack, so the pack will not fit in one jar. Repack to one jar fill or less before the next episode.
- **Defective bulk pack:** a torn or already open pack, or one whose rolled top will not unroll. Replace it before the next episode.
- **Defective jar or lid:** a cracked jar, a worn off fill line, or a thread that will not take a lid. Replace it before the next episode.
- **Defective label:** a label with no adhesive left, writing that cannot be read, or a label that tears or folds onto itself during the peel. Replace the sheet before the next episode.
- **Pack and label do not match:** the label on the sheet names a product other than the one in the pack, so no label matches the product decanted. Restage before the next episode.
- **Crate has no open slot** when the episode begins, so the finished jar has nowhere to stand. Empty the crate before the next episode.
- **Jar slides** under a correct steady hold because the table is slick or crumbed. Wipe and dry the table before the next episode.

Annotation subtasks (from SOP)

1. Stage the jar and the pack in the working area
2. Open the bulk pack
3. Decant the pack into the jar
4. Discard the empty pack
5. Close the jar with its lid
6. Apply the written label to the jar
7. Shelf the closed and labeled jar in the crate
8. Confirm the jar and end the episode